

# **FAST ADDER USING LOGICAL EFFORT METHOD**

*By*

|                  |           |
|------------------|-----------|
| AMIRTHA PRASAD   | 22BEC1002 |
| SHERINA ESTHER D | 22BEC1198 |
| DEEPIKA B        | 22BEC1367 |
| V. SRISHTI       | 22BEC1517 |

A report submitted to

**Dr. S. Umadevi**

**SCHOOL OF ELECTRONICS ENGINEERING**

in partial fulfillment of the requirements for the course

**BECE303L VLSI System Design**

in

**B.Tech ELECTRONICS AND COMMUNICATION ENGINEERING**



Vandalur– Kelambakkam Road

Chennai– 600127

*November 2024*

## **CERTIFICATE**

*This is to certify that the Project work titled “**FAST ADDER USING LOGICAL EFFORT METHOD**” is being submitted by Amirtha Prasad- 22BEC1002, Sherina Esther D- 22BEC1198, Deepika B- 22BEC1367 & V. Srishti- 22BEC1517 for the course BECE303L VLSI Systems Design is a record of bonafide work done under my guidance. The contents of this project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University.*

Professor

Dr. S. Umadevi

## **ABSTRACT**

In modern digital circuits, the propagation delay of arithmetic units is a critical factor influencing the overall system performance. Adders, being key components in arithmetic operations, must be designed for minimal delay to meet the demands of high-speed applications. This project investigates the design of a Fast Adder using the Logical Effort Method, focusing exclusively on reducing propagation delay, and compares its performance with the conventional Ripple Carry Adder (RCA).

The Ripple Carry Adder, while simple and widely used, suffers from high propagation delay due to the sequential carry propagation through all its stages. In contrast, the Logical Effort Method provides a systematic approach to minimize delay by optimizing gate sizing and balancing logical, electrical, and parasitic efforts.

Key aspects of this project include:

1. Logical Effort-Based Design: Applying the Logical Effort Method to design a fast adder, such as a Carry Lookahead Adder (CLA), optimized for minimal propagation delay.
2. Delay Analysis: Theoretical and practical evaluation of propagation delay for both the optimized fast adder and the Ripple Carry Adder.
3. Simulation and Validation: Performance benchmarking using simulation tools to validate delay improvements achieved by the proposed design.

This project highlights the significant reduction in propagation delay achieved through the Logical Effort Method compared to the Ripple Carry Adder. The findings emphasize the importance of advanced design methodologies in developing high-speed and efficient digital systems for modern applications.

## ACKNOWLEDGEMENT

We wish to express our sincere thanks and deep sense of gratitude to our project guide, **Dr.S. Umadevi**, School of Electronics Engineering, for his consistent encouragement and valuable guidance offered to us in a pleasant manner throughout the course of the project work.

We are extremely grateful to **Dr. A. Ravi Sankar**, Dean of School of Electronics Engineering, VIT Chennai, for extending the facilities of the school towards our project and for his unstinting support.

We express our thanks to our Head of the Department **Dr. K. Mohanaprasad** for his support throughout the course of this project.

We also take this opportunity to thank all the faculty of the school for their support and their wisdom imparted to us throughout the course.

We thank our parents, family, and friends for bearing with us throughout the course of our project and for the opportunity they provided us in undergoing this course in such a prestigious institution.

## CONTENTS PAGE

| SNO | TITLE                  | PAGE NO |
|-----|------------------------|---------|
| 1   | ABSTRACT               | 3       |
| 2   | ACKNOWLEDGEMENT        | 4       |
| 3   | INTRODUCTION           | 6       |
| 4   | LITERATURE SURVEY      | 7-8     |
| 5   | METHODOLOGY            | 9-12    |
| 6   | RESULTS AND DISCUSSION | 13-17   |
| 7   | INFERENCES             | 18      |
| 8   | CONCLUSION             | 19      |
| 9   | REFERENCES             | 20      |

# INTRODUCTION

In digital systems, adders are fundamental components used in a variety of arithmetic operations, such as addition, subtraction, multiplication, and division. Their performance directly impacts the overall speed of processors and other computational units. As modern applications demand faster and more efficient systems, the need for high-speed adder designs becomes increasingly important.

The **Ripple Carry Adder (RCA)**, one of the simplest adder architectures, is known for its straightforward design and minimal area requirement. However, it suffers from high propagation delay because the carry signal must sequentially propagate through each stage of the adder. This limitation makes it unsuitable for high-speed applications.

To overcome this, advanced design techniques like the **Logical Effort Method** are employed to optimize the performance of adders. The Logical Effort Method is a systematic approach that focuses on minimizing propagation delay by balancing the logical effort, electrical effort, and parasitic delay of the gates in a circuit. It provides a framework for selecting optimal gate sizes and designing circuits that achieve the best possible speed for a given logic function.

This project aims to design a **fast adder** using the Logical Effort Method and compare its propagation delay performance with that of the Ripple Carry Adder. By leveraging this method, advanced adder architectures such as the **Carry Lookahead Adder (CLA)** or other optimized designs are implemented to significantly reduce delay.

The objectives of this project are:

1. To understand the limitations of the RCA in terms of propagation delay.
2. To design an optimized fast adder using the Logical Effort Method.
3. To compare the propagation delay of the proposed fast adder with that of the RCA through theoretical calculations and simulations.

The results of this project will demonstrate the effectiveness of the Logical Effort Method in enhancing adder performance, providing valuable insights for the design of high-speed arithmetic units in modern digital systems.

# LITERATURE SURVEY

## BASE PAPER

[https://www.researchgate.net/publication/372472109\\_Design\\_and\\_analysis\\_of\\_a\\_novel\\_fast\\_adder\\_using\\_logical\\_effort\\_method](https://www.researchgate.net/publication/372472109_Design_and_analysis_of_a_novel_fast_adder_using_logical_effort_method)

The paper introduces a novel fast adder design optimized using the Logical Effort Method to improve propagation delay in digital systems. The Ripple Carry Adder (RCA) is widely used for its simplicity but suffers from high propagation delay due to the sequential carry propagation through all stages. The authors propose a new architecture that minimizes this delay by systematically optimizing the circuit's structure and gate sizing using the Logical Effort Method.

The Logical Effort Method involves balancing three key factors: logical effort (the inherent delay of a gate), electrical effort (the ratio of load to input capacitance), and parasitic delay (intrinsic delay from internal gate capacitances). By applying this method, the proposed adder design achieves a balanced distribution of delay across the circuit paths, significantly reducing overall computation time. The design is implemented using multiplexers and OR gates, ensuring modularity and scalability to larger bit-widths.

The proposed adder's performance is compared with several traditional adder architectures, including the RCA, Carry Lookahead Adder (CLA), Carry Skip Adder, Carry Select Adder (CSLA), and Kogge-Stone Adder. The results demonstrate that the new design achieves a substantial reduction in propagation delay, especially for larger bit-widths like 32-bit and 64-bit configurations. It also shows improvements in power-delay product (PDP) and delay-area product, making it highly efficient for high-performance applications.

The study concludes that the novel adder provides significant advantages over conventional designs, particularly in terms of speed and efficiency, and is a robust solution for modern digital systems requiring fast computation.

## Key Contributions

### 1. Novel Adder Architecture:

- The proposed adder introduces a **modular and hierarchical structure** suitable for high-speed computation.
- It minimizes the number of logic levels in the circuit, reducing the overall propagation delay.

### 2. Logical Effort Methodology:

- The Logical Effort Method is used to determine the **optimal gate sizing** and configurations for minimum delay.
- Key aspects of this methodology include:

- **Logical Effort (g):** Quantifies the gate's ability to drive the load compared to an ideal inverter.
- **Electrical Effort (h):** The ratio of output load to the gate input capacitance.
- **Parasitic Delay (p):** Intrinsic delay caused by the gate itself.
- By analyzing these factors, the adder design ensures balanced delays across different paths in the circuit.

### 3. Implementation Details:

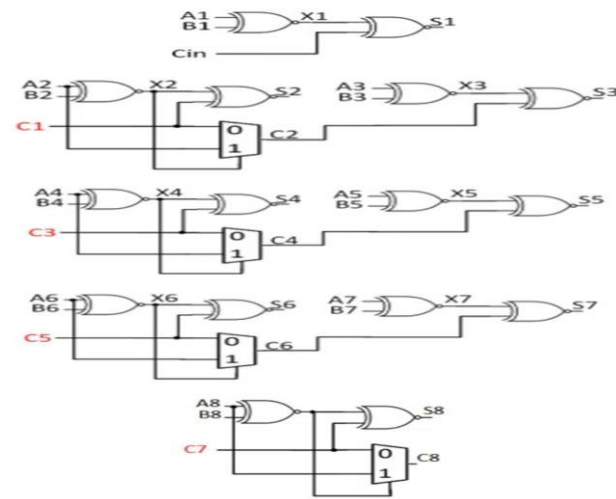
- The adder is designed using **multiplexers and OR gates** for faster carry propagation.
- A **layered structure** is adopted to improve scalability to larger bit-widths (e.g., 8, 16, 32, and 64 bits).
- Mathematical models are developed to calculate delay and other performance metrics for each layer of the adder.

### 4. Comparative Evaluation:

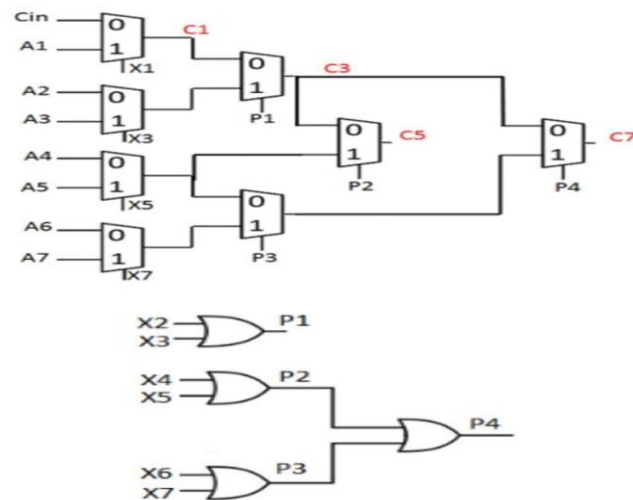
- The performance of the proposed adder is compared with traditional adder designs, including:
  - **Ripple Carry Adder (RCA):** Simple but slow due to sequential carry propagation.
  - **Carry Lookahead Adder (CLA):** Faster than RCA but more complex.
  - **Carry Skip Adder (CSA):** Reduces carry propagation delay with bypass paths.
  - **Carry Select Adder (CSLA):** Improves speed but at the cost of increased area.
  - **Kogge-Stone Adder:** A parallel-prefix adder known for high speed but with higher area and power overheads.
- The proposed adder achieves a significant reduction in delay compared to these architectures.



## METHODOLOGY



### SUM FOR 8 BIT FAST ADDER



### CARRY FOR 8 BIT FAST ADDER

The methodology for designing and analyzing the novel fast adder using the Logical Effort Method is outlined in the paper through a detailed and systematic approach. Here is a more elaborate explanation of the process:

#### 1. Application of the Logical Effort Method

The Logical Effort Method is central to the design, providing a systematic way to minimize propagation delay in digital circuits. The methodology involves three key factors:

- **Logical Effort (g):**

This quantifies the delay introduced by the logic gate itself when driving a load compared to an ideal inverter. Each gate type (e.g., AND, OR, MUX) has a specific logical effort value that is used to determine its contribution to the overall delay.

- **Electrical Effort (h):**

Electrical effort is the ratio of the output load to the input capacitance of the gate. This factor helps balance the load between successive stages in the circuit, ensuring uniform performance.

- **Parasitic Delay (p):**

This is the intrinsic delay of a gate caused by internal capacitances. It depends on the physical structure of the gate and contributes to the delay regardless of the load.

The method focuses on balancing these three factors across all stages of the adder to minimize delay. Each stage is designed so that the product of these factors—referred to as the stage effort—is constant throughout the circuit, ensuring the optimal propagation time.

## 2. Adder Architecture and Design

The proposed adder architecture departs from the conventional Ripple Carry Adder (RCA) design, which suffers from high propagation delay due to sequential carry propagation. The methodology incorporates the following design principles:

- **Use of Multiplexers and OR Gates:**

The design simplifies the carry propagation mechanism by utilizing multiplexers and OR gates to reduce the number of logic levels. This minimizes the carry propagation delay, especially in high-bit-width adders.

- **Modular and Hierarchical Structure:**

The adder is designed in a modular manner, making it scalable for higher bit-widths like 8-bit, 16-bit, 32-bit, and 64-bit. The hierarchical structure allows

efficient implementation without a significant increase in delay as the number of bits increases.

- **Minimizing Critical Path Delay:**

The critical path, which determines the total propagation delay, is carefully optimized by ensuring that delays in all paths are well-balanced. This is achieved by proper sizing of gates and selecting optimal configurations using the Logical Effort Method.

### 3. Implementation and Mathematical Modeling

- **Delay Equations:**

Mathematical models for delay are derived for each stage of the adder. These equations incorporate logical effort, electrical effort, and parasitic delay, allowing the calculation of the optimal gate sizes and the overall delay for the circuit.

- **Gate Sizing Optimization:**

Using the delay equations, gate sizes are adjusted to achieve the lowest possible propagation delay while maintaining reasonable power and area overhead.

- **Scalability:**

The modular design ensures that the same optimization principles can be applied to larger bit-width adders, with the delay increasing sub-linearly as the bit-width grows.

### 4. Comparative Analysis

The proposed adder is compared against several traditional adder architectures:

- **Ripple Carry Adder (RCA):**

Used as a baseline for comparison, it exhibits the highest propagation delay due to sequential carry propagation.

- **Carry Lookahead Adder (CLA):**

Reduces carry propagation delay using pre-computation but involves complex logic.

- **Carry Skip Adder (CSA):**

Incorporates bypass paths to skip carry propagation, achieving moderate delay reduction.

- **Carry Select Adder (CSLA):**

Pre-computes potential carry outputs for reduced delay but at the cost of increased area.

- **Kogge-Stone Adder:**

A parallel-prefix adder with high speed but significant area and power requirements.

Simulations and theoretical calculations for 8-bit, 16-bit, 32-bit, and 64-bit configurations demonstrate the superiority of the proposed design in terms of propagation delay, especially for larger bit-widths.

## 5. Evaluation Metrics

Performance is evaluated based on the following metrics:

- **Propagation Delay:**

The time required for the signal to propagate through the circuit, with the proposed design achieving significant reductions compared to conventional adders.

- **Power-Delay Product (PDP):**

A measure of energy efficiency, where the proposed design exhibits lower values, indicating better performance.

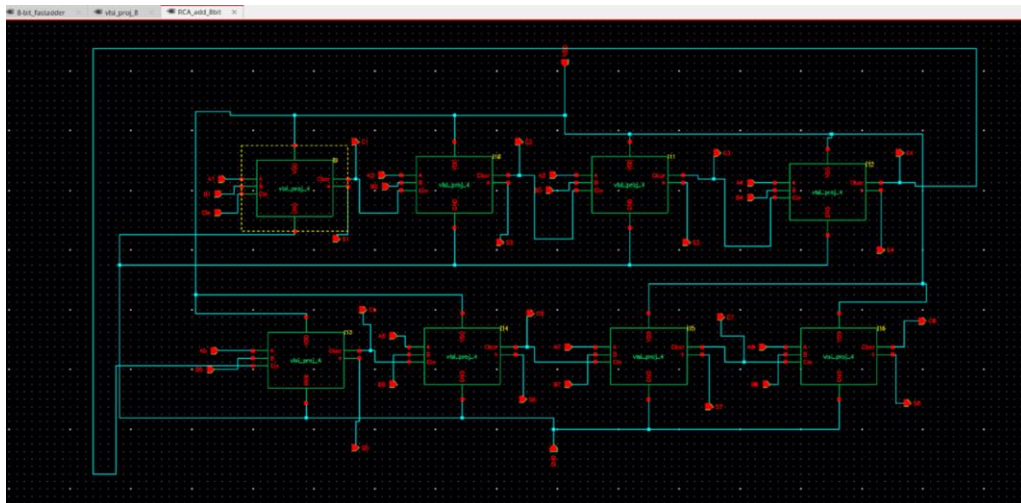
- **Delay-Area Product (DAP):**

A combined measure of speed and silicon area efficiency, with the proposed adder striking a balance between the two.

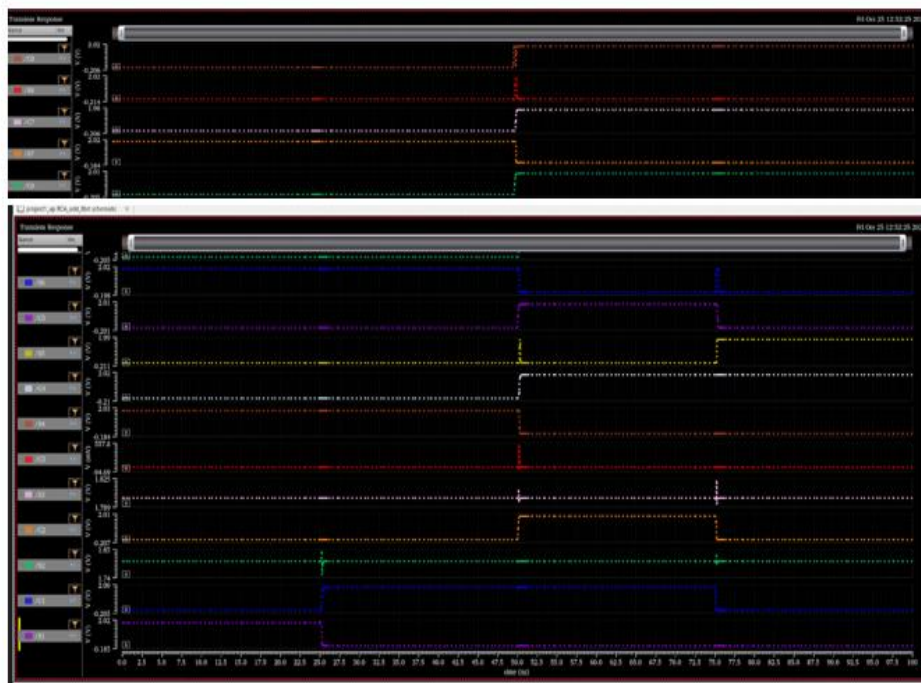
# RESULTS AND DISCUSSION

## SCREENSHOTS

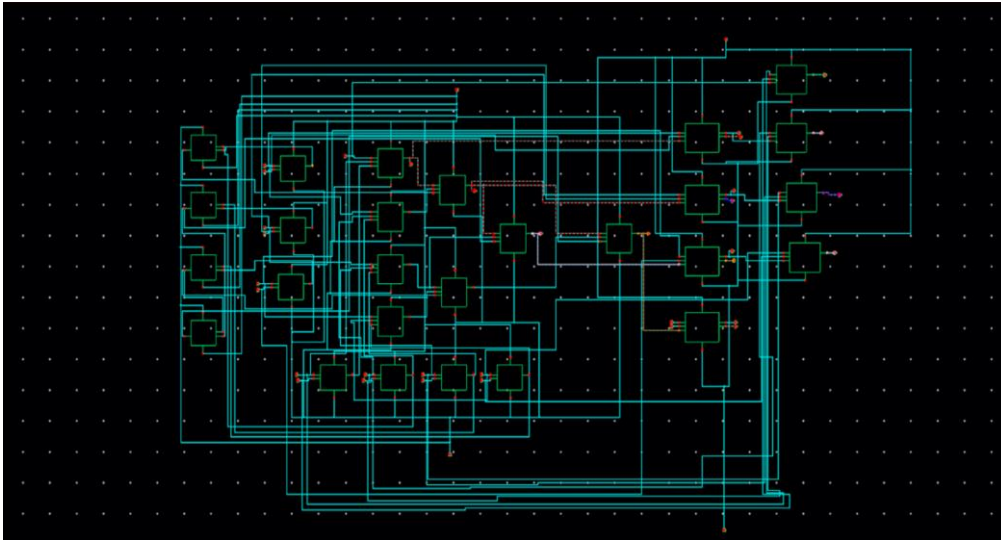
### 1. RCA 8 BIT ADDER CIRCUIT:



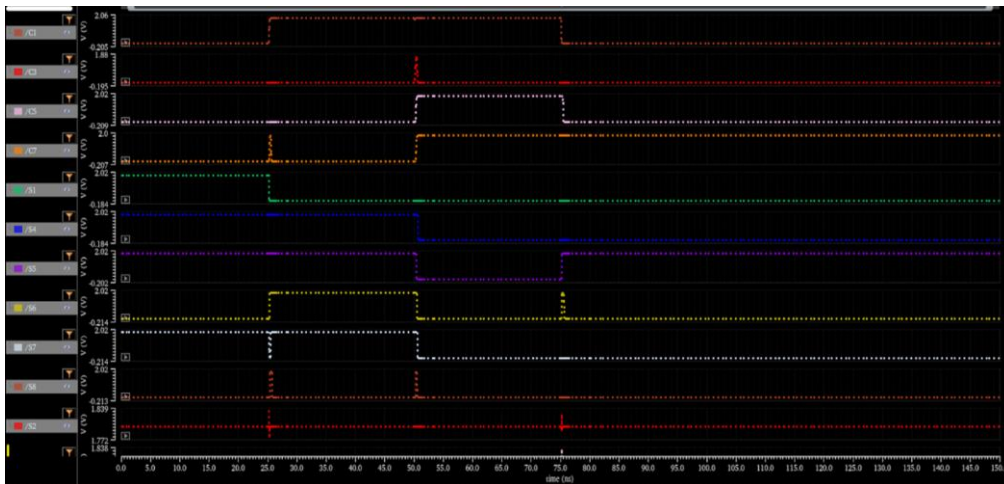
### 2. RCA GRAPH:



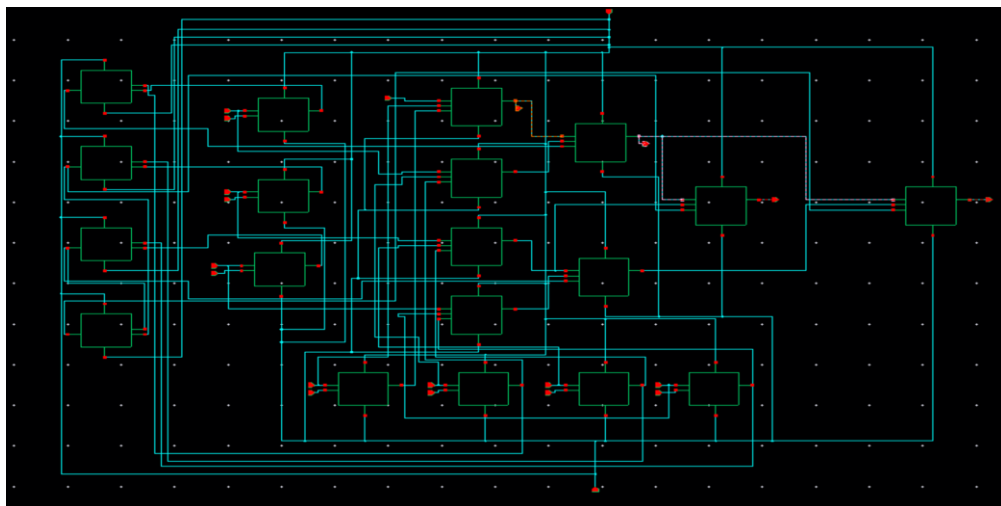
### 3. 8 BIT FAST ADDER SUM CIRCUIT:



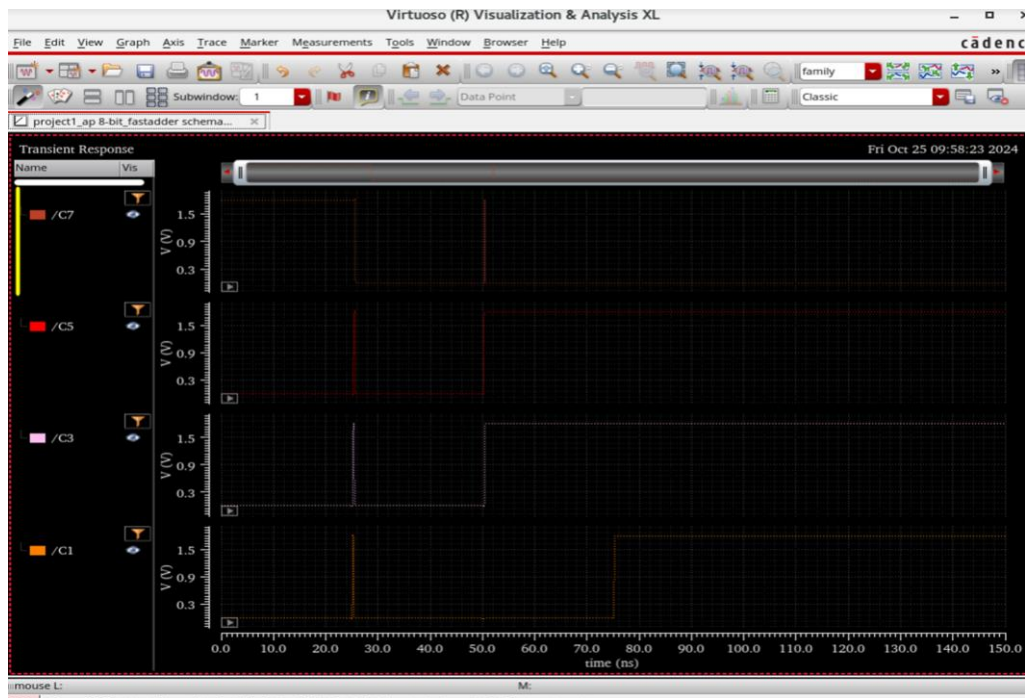
### 4. FAST ADDER SUM GRAPH:



### 5. 8 BIT FAST ADDER CARRY CIRCUIT:



## 6. FAST ADDER CARRY GRAPH:



## DELAY CALCULATION AND COMPARISON

| A        | B        | Cin | Input  | Output | rise/fall | RCA(ps) | Fast Adder(ps) |
|----------|----------|-----|--------|--------|-----------|---------|----------------|
| 01101001 | 00010101 | 0   | A6(1r) | S7     | rise      | 298.1   | 274.8          |
| 10011011 | 01101011 | 0   | A6(1f) | S7     | fall      | 280.4   | 276.7          |
| 10101000 | 01101110 | 0   | B2(1r) | S3     | rise      | 305.3   | 279.4          |
| 01101001 | 00010001 | 0   | B2(1f) | S3     | fall      | 309.2   | 283.9          |
| 10011011 | 01101011 | 0   | A5(1r) | C5     | rise      | 254.6   | 238.5          |
| 10101000 | 01101110 | 0   | B3(1r) | C5     | fall      | 231.8   | 217.6          |
| 10011011 | 01101011 | 0   | B6(1r) | C6     | rise      | 269.9   | 244.7          |
| 01001001 | 00010110 | 0   | A6(2f) | C6     | fall      | 247.5   | 223.1          |

## AVERAGE DELAY

|              | <b>RCA</b>   | <b>Fast Adder</b> |
|--------------|--------------|-------------------|
| <b>SUM</b>   | RISE: 301.7  | RISE: 277.1       |
|              | FALL: 294.8  | FALL: 280.3       |
| <b>CARRY</b> | RISE: 262.25 | RISE: 241.6       |
|              | FALL: 239.65 | FALL: 220.35      |

In the context of digital circuits, specifically for the project comparing a FAST ADDER using the Logical Effort method and a Ripple Carry Adder (RCA), we are focusing on the delay comparison between these two approaches for implementing an adder.

### 1. Ripple Carry Adder (RCA):

An RCA is the simplest form of adder, where each bit of the sum is computed sequentially. In an  $n$ -bit RCA, each full adder computes the sum of two bits along with a carry-in from the previous stage, and generates a carry-out that is passed to the next stage.

- **Delay of RCA:** The total delay of an RCA is dominated by the worst-case delay of the longest carry propagation path, which is  $O(n)$ , where  $n$  is the number of bits. Specifically, the delay is determined by the time it takes for the carry to ripple through all the stages.
  - Each full adder has a delay due to the logic gates involved (AND, OR, XOR) and the propagation of the carry signal.
  - As each bit addition depends on the carry from the previous bit, the overall delay increases linearly with the number of bits.

### 2. Fast Adder using Logical Effort Method:

The Logical Effort method is a technique to design and optimize the delay of digital circuits by balancing the size of transistors and the number of gates involved in each stage. The goal is to minimize the total delay by controlling the delay through proper sizing of the transistors (and thus the capacitance).



In the case of a Fast Adder using this method, typically the adder is designed to minimize the critical path delay, which is the time it takes for a signal to propagate from one input to the final output. Logical effort seeks to balance the stages in such a way that the delay is minimized and more uniform across all stages.

- **Delay of Fast Adder (using Logical Effort):** The delay of a fast adder designed with the Logical Effort method is generally  $O(\log n)$  for  $n$  bit addition. This is because it aims to reduce the carry propagation problem by using carry-lookahead or other techniques like carry-select, carry-skip, or Kogge-Stone adders.
  - These methods involve more complex logic structures than RCA, allowing faster computation because they avoid the ripple effect in the carry chain, effectively reducing the overall delay.

### 3. Comparison of Delay:

When comparing the delays between the Ripple Carry Adder (RCA) and the Fast Adder (using Logical Effort), the key differences are:

- **RCA Delay:** The delay is linear in terms of the number of bits, i.e.,  $O(n)$ . For larger values of  $n$ , this becomes a significant bottleneck, especially for applications requiring fast arithmetic operations.
- **Fast Adder Delay (Logical Effort):** The delay is logarithmic in terms of the number of bits, i.e.,  $O(\log n)$ . As  $n$  increases, the difference in performance becomes more significant, as the delay of a fast adder grows much slower compared to the RCA.

## INFERENCE

The paper presents a novel adder design that utilizes the Logical Effort Method to optimize performance in high-speed digital systems. The Logical Effort Method, known for its systematic approach to optimizing circuit design, allows for the precise tuning of the adder's logical effort, electrical effort, and parasitic delays. This results in a significant reduction in propagation delay, making the proposed adder highly suitable for high-performance applications where speed is critical. By balancing these key factors, the adder achieves a design that minimizes the critical path delay while maintaining scalability for larger bit widths, such as 32-bit and 64-bit configurations. One of the main strengths of this new adder is its superior performance when compared to conventional adder architectures, including the Ripple Carry Adder (RCA), Carry Lookahead Adder (CLA), and Kogge-Stone Adder. Traditional adders, while efficient for smaller bit widths, often face challenges as bit-width increases, leading to higher propagation delays. The proposed design, by applying the Logical Effort Method, mitigates these delays, particularly when scaling up to larger bit configurations. This makes the proposed adder particularly beneficial for modern processors and data-intensive digital systems, where large bit-widths are often required. In addition to reduced propagation delay, the proposed adder also demonstrates a favorable power-delay product (PDP) and delay-area product (DAP).

The power-delay product measures the energy consumption of the circuit relative to its speed, and a lower PDP indicates better efficiency in both speed and power usage. The delay-area product, on the other hand, quantifies the tradeoff between circuit delay and the area occupied by the adder. By optimizing these parameters, the proposed design offers not only enhanced speed but also energy efficiency, a critical factor in modern computing systems where power consumption is a major concern. The study further emphasizes the robustness of the Logical Effort Method as a circuit design approach. Through its systematic optimization process, this method provides a powerful tool for achieving optimal performance in digital systems. The results of this study confirm that the proposed adder is not only faster but also more energy-efficient than traditional adders. This combination of reduced delay and energy efficiency makes it an ideal candidate for high-performance applications such as processors, memory units, and other data-intensive digital systems where both low propagation delay and power efficiency are paramount. In conclusion, the proposed adder design offers substantial improvements in terms of speed, energy efficiency, and scalability. The paper highlights the practical advantages of using the Logical Effort Method for optimizing critical path delay and balancing tradeoffs between speed, power consumption, and area. As digital systems continue to evolve and demand higher performance, the insights provided by this study will be invaluable for developing efficient, high-speed adders suitable for modern processors and other high-performance digital circuits.

## CONCLUSION

The paper presents a novel adder design that utilizes the Logical Effort Method to optimize performance in high-speed digital systems. The Logical Effort Method, known for its systematic approach to optimizing circuit design, allows for the precise tuning of the adder's logical effort, electrical effort, and parasitic delays. This results in a significant reduction in propagation delay, making the proposed adder highly suitable for high-performance applications where speed is critical. By balancing these key factors, the adder achieves a design that minimizes the critical path delay while maintaining scalability for larger bit-widths, such as 32-bit and 64-bit configurations.

One of the main strengths of this new adder is its superior performance when compared to conventional adder architectures, including the Ripple Carry Adder (RCA), Carry Lookahead Adder (CLA), and Kogge-Stone Adder. Traditional adders, while efficient for smaller bit-widths, often face challenges as bit-width increases, leading to higher propagation delays. The proposed design, by applying the Logical Effort Method, mitigates these delays, particularly when scaling up to larger bit configurations. This makes the proposed adder particularly beneficial for modern processors and data-intensive digital systems, where large bit-widths are often required.

In addition to reduced propagation delay, the proposed adder also demonstrates a favorable power-delay product (PDP) and delay-area product (DAP). The power-delay product measures the energy consumption of the circuit relative to its speed, and a lower PDP indicates better efficiency in both speed and power usage. The delay-area product, on the other hand, quantifies the tradeoff between circuit delay and the area occupied by the adder. By optimizing these parameters, the proposed design offers not only enhanced speed but also energy efficiency, a critical factor in modern computing systems where power consumption is a major concern.

In conclusion, the proposed adder design offers substantial improvements in terms of speed, energy efficiency, and scalability. The paper highlights the practical advantages of using the Logical Effort Method for optimizing critical path delay and balancing tradeoffs between speed, power consumption, and area. As digital systems continue to evolve and demand higher performance, the insights provided by this study will be invaluable for developing efficient, high-speed adders suitable for modern processors and other high-performance digital circuits.

## REFERENCES

- [https://www.researchgate.net/publication/372472109\\_Design\\_and\\_analysis\\_of\\_a\\_novel\\_fast\\_adder\\_using\\_logical\\_effort\\_method](https://www.researchgate.net/publication/372472109_Design_and_analysis_of_a_novel_fast_adder_using_logical_effort_method)
- [https://www.researchgate.net/publication/383164150\\_An\\_Approach\\_to\\_Design\\_a\\_Low\\_Power\\_High-Speed\\_Full\\_Adder\\_Circuit\\_Based\\_on\\_Logical\\_Effort](https://www.researchgate.net/publication/383164150_An_Approach_to_Design_a_Low_Power_High-Speed_Full_Adder_Circuit_Based_on_Logical_Effort)
- [https://www.researchgate.net/publication/345184288\\_Design\\_and\\_Analysis\\_of\\_High-Speed\\_Parallel\\_Prefix\\_Adder\\_for\\_Digital\\_Circuit\\_Design\\_Applications](https://www.researchgate.net/publication/345184288_Design_and_Analysis_of_High-Speed_Parallel_Prefix_Adder_for_Digital_Circuit_Design_Applications)
- [https://www.researchgate.net/publication/367545192\\_Design\\_and\\_Implementation\\_of\\_an\\_Efficient\\_Multiplier\\_for\\_Fir\\_Filter](https://www.researchgate.net/publication/367545192_Design_and_Implementation_of_an_Efficient_Multiplier_for_Fir_Filter)
- [https://www.researchgate.net/publication/3938391\\_Application\\_of\\_logical\\_effort\\_on\\_delay\\_analysis\\_of\\_64-bit\\_static\\_carry-lookahead\\_adder](https://www.researchgate.net/publication/3938391_Application_of_logical_effort_on_delay_analysis_of_64-bit_static_carry-lookahead_adder)